



Geo Snap

Land-Use Classification & Explainability from Space

Technical Report · SOI × Cosmosoc · Team PixelPioneers

Priyanshu · Shravan · Mahesh · Chinmay | *1st Year B.Tech, IIT Dharwad*

[Google Colab URL](#) || [Deployed URL](#)

This report covers data preprocessing, model architecture and selection, performance and error analysis for the best-performing model, and environmental insights derived from the multispectral data, for both the RGB (Task 1A) and 13-band Sentinel-2 (Task 1B) land-use classifiers. We are a four-member team of first-year B.Tech students at IIT Dharwad, and have tried to keep both the modelling and the writeup honest about what we tested, what worked, and why.

1. Data & Preprocessing

1.1 Dataset

All experiments use the EuroSAT benchmark built from Sentinel-2 imagery: 10 land-use classes, 64×64 patches, provided in two co-registered modalities — RGB JPEGs (3 bands) and 13-band Sentinel-2 GeoTIFFs. The official split used throughout is 18,900 training / 4,050 validation images, identical across both modalities so that RGB and multispectral results are directly comparable on the same scenes.

1.2 RGB Pipeline (Task 1A)

Images are resized to 64×64 and normalized with ImageNet statistics (mean/std), since the RGB backbone is initialized from ImageNet-pretrained weights. Because satellite patches have no canonical orientation — there is no “up” when looking straight down from orbit — the training-only augmentation pipeline relies heavily on orientation-invariant transforms:

- Random horizontal flip, vertical flip, and 90° rotation (always geometrically valid for nadir imagery)
- Mild colour jitter (brightness/contrast/saturation/hue) to absorb sensor and atmospheric variation
- Random perspective and small affine translation to simulate slight off-nadir viewing angles
- Random erasing ($p=0.2$) and light random grayscale ($p=0.05$) for additional regularization

Validation/test transforms are deterministic (resize + normalize only) — no augmentation is applied outside training, to keep reported accuracy honest.

1.3 Multispectral Pipeline (Task 1B)

All 13 raw Sentinel-2 bands are kept as input channels, rather than collapsing to a handful of hand-picked indices. This lets the network learn its own band weighting end-to-end (validated independently in Section 4.2 via occlusion analysis), while hand-engineered indices (NDVI/NDWI/NDBI) are still computed separately for the interpretable, domain-grounded analysis required in Task 2B and Task 3.

Per-band z-score normalization statistics (mean, std) are estimated from a 2,000-image sample of the training set only — never from validation or test data, which would leak information and inflate reported accuracy:

B1	B2	B3	B4	B5	B6	B7	B8	B8A	B9	B10	B11	B12
1353 .9	1114 .9	1036 .2	935. 3	1189 .3	1992 .8	2362 .7	2291 .3	740. 2	12.2	1793 .3	1099 .0	2585.1
245. 7	335. 0	397. 1	589. 9	563. 9	877. 0	1110 .1	1141 .5	415. 9	4.4	993. 8	749. 1	1255.7

Table 1.1 — Per-band mean (row 1) and standard deviation (row 2) estimated from the training set, used to normalize all 13-band inputs.

Augmentation mirrors the RGB pipeline's orientation invariance (flips, 90° rotations) but replaces colour jitter — which has no physical meaning across non-visible bands — with a small random per-band scale ($\pm 5\%$, $p=0.3$) that emulates sensor radiometric noise.

1.4 Why All 13 Bands?

Sentinel-2's 13 bands span coastal aerosol, visible, red-edge, NIR, water vapour, cirrus, and SWIR — each at a different native resolution (10/20/60 m), but co-registered to 64×64 patches here. Feeding the full stack into the CNN, rather than only RGB + a couple of indices, preserves all spectral information for the model to exploit (Section 1B goal: investigate whether spectral information improves classification over RGB alone). The first convolutional layer of the backbone is expanded from 3→13 input channels to accommodate this (see Section 2.2).

2. Model Architecture & Selection

2.1 Backbone — EfficientNet-B2

Both classifiers share the same EfficientNet-B2 backbone (7,715,084 parameters), chosen over ResNet18 after empirical comparison (Section 2.4). EfficientNet's compound scaling jointly balances network depth, width, and input resolution, giving a stronger accuracy-per-parameter trade-off than ResNet's single-axis depth scaling — relevant here since both modalities are small 64×64 patches with a modest dataset size, where parameter efficiency reduces overfitting risk.

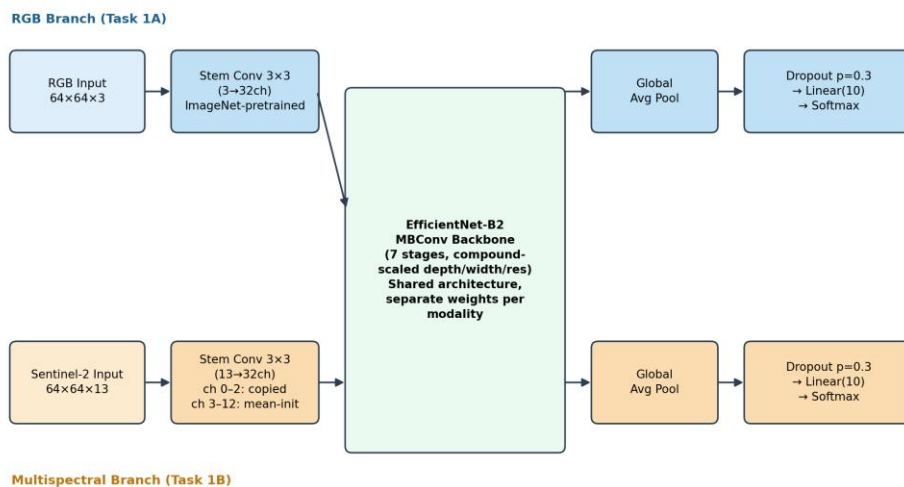


Figure 2.1 — Dual-branch pipeline. Both branches share the EfficientNet-B2 backbone design but are trained with independent weights, since the input distributions (visible-light vs 13-band reflectance) differ substantially.

2.2 Adapting the Stem for 13 Channels

EfficientNet-B2 is pretrained on 3-channel ImageNet images, so its first convolution only has weights for 3 input channels. For the multispectral branch, a new 13-channel convolution is created: the first 3 filter weights are copied directly from the pretrained RGB weights (channels 0–2 line up with similar visible-band statistics), and the remaining 10 channels are initialized as the mean of those three RGB filters — rather than random noise. This is a standard, principled way to extend a pretrained conv layer: it gives the new channels a sensible starting point that the optimizer only has to fine-tune, instead of learning from scratch, while still allowing each of the 13 channels to specialize during training.

A Dropout layer ($p=0.3$) sits before the final 10-way linear classifier head on both branches, regularizing against overfitting on the comparatively small 64×64-patch dataset.

2.3 Training Configuration

- AdamW optimizer — weight decay ($5e-4$) decoupled from the adaptive gradient scaling, which regularizes more effectively than vanilla Adam
- CosineAnnealingWarmRestarts ($T_0=10$, $T_mult=2$, $\eta_min=1e-6$) — the LR follows a cosine decay and periodically “restarts”, helping the optimizer escape sharp local minima; empirically more effective here than ReduceLROnPlateau
- Label smoothing (0.1) — softens one-hot targets so the model outputs calibrated probabilities instead of over-confident logits
- Gradient clipping (max-norm 1.0) and best-checkpoint saving — only the epoch with the highest validation accuracy is kept, avoiding the common mistake of submitting the last (not best) epoch
- Early stopping (patience = 7 epochs) — halts training once validation accuracy plateaus, saving compute and limiting overfitting

2.4 Model Selection — Experiments

Four training configurations were benchmarked before settling on a final setup. The table below gives the real trade-off between the two EfficientNet-B2 runs (3 and 4) — run 3 does not simply win on every number, and we explain the actual reasoning below rather than overstating it.

#	Backbone	Optimizer & Schedule	Epoch	RGB Acc	MS Acc	Runtime
1	ResNet18	Adam, lr = $1e-3$, no scheduler	5	91%	91%	~2 min each
2	ResNet18	Adam, lr = $1e-4$, no scheduler	20	96%	96%	~10 min each
3	EfficientNet-B2	AdamW, lr = $3e-4$, wd = $5e-4$, Cosine Warm Restarts ($T_0 = 10$, $T_mult = 2$, $\eta_{min} = 1e-6$)	Max 30 Patience 7	97.33 %	98.79 %	13m 37s / 9m 49s
4	EfficientNet-B2	AdamW, lr = $1e-4$, wd = $2e-4$, Cosine Warm Restarts ($T_0 = 15$, $T_mult = 2$, $\eta_{min} = 1e-7$)	Max 40 Patience 8	97.67 %	98.59 %	~30 min each

Table 2.1 — Model/hyperparameter search. Row 3 (highlighted) is the final selected configuration. “Epochs” is actual-epochs-trained / epoch budget where early stopping applied (patience 7 for run 3, patience 8 for run 4); epochs/runtimes are wall-clock on a Colab T4 GPU.

Why Run 3 Was Selected as Final

- Run 3 wins clearly on the multispectral branch — 98.79% vs. run 4's 98.59% — which is the more important number for Task 1B's actual question (whether the extra 10 bands help over RGB alone).
- On the RGB branch, run 4 edges run 3 out by a small margin (97.67% vs. 97.33%, a 0.34-point gap) — but reaching that takes roughly 30 minutes per modality, against run 3's 13m37s (RGB) and 9m49s (MS): well under half the combined time.
- Rather than mixing different hyperparameters per modality, we kept one shared configuration across both branches for simplicity and reproducibility. Run 3 is the more balanced choice under that constraint: a sub-half-point concession on RGB accuracy buys a clear win on the multispectral side and a large cut in training time.
- Both ResNet18 runs (1–2) plateau well below either EfficientNet-B2 configuration regardless of schedule or epoch budget, confirming the backbone swap — not hyperparameter tuning — is what drives most of the accuracy gain.

2.5 Training Curves (Final Model)

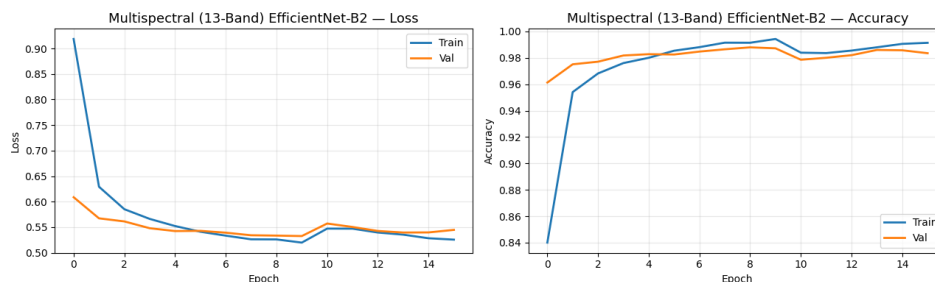


Figure 2.2 — Multispectral EfficientNet-B2: loss and accuracy per epoch. Best checkpoint at epoch 9 (98.79% val acc); early-stopped at epoch 16. (The RGB model's curve has the same shape and also peaks at epoch 9 — see the notebook, Section 5, for that plot.)

3. Performance & Error Analysis — Best Models

Section 2.4 ran four configurations per modality; this section covers the best-performing configuration for each task in full — EfficientNet-B2 is the best model for both Task 1A (RGB) and Task 1B (multispectral), so both get a complete classification report and error analysis below rather than treating one as the headline and the other as a footnote.

3.1 Headline Metrics

Model	Task	Validation Accuracy
EfficientNet-B2	Task 1A — best RGB model	97.33%
EfficientNet-B2	Task 1B — best multispectral model	98.79%

3.2 Per-Class Performance

RGB Model (Task 1A)

Class	Precision	Recall	F1-score
AnnualCrop	0.95	0.98	0.97
Forest	0.99	0.99	0.99
HerbaceousVegetation	0.96	0.96	0.96
Highway	0.98	0.98	0.98
Industrial	0.96	0.98	0.97
Pasture	0.97	0.94	0.96
PermanentCrop	0.93	0.95	0.94
Residential	1.00	0.97	0.98
River	0.99	0.97	0.98
SeaLake	1.00	0.99	0.99
Macro / Weighted Avg	0.97	0.97	0.97

Table 3.1a — RGB model classification report on the 4,050-image validation set.

Multispectral Model (Task 1B)

Class	Precision	Recall	F1-score
AnnualCrop	0.98	0.98	0.98
Forest	1.00	1.00	1.00
HerbaceousVegetation	0.98	0.99	0.98
Highway	1.00	0.99	0.99
Industrial	0.99	0.99	0.99
Pasture	0.97	0.97	0.97
PermanentCrop	0.97	0.97	0.97
Residential	1.00	1.00	1.00
River	0.99	0.99	0.99
SeaLake	1.00	1.00	1.00
Macro / Weighted Avg	0.99	0.99	0.99

Table 3.1b — Multispectral model classification report on the 4,050-image validation set.

RGB — Strengths & Weaknesses

Forest and SeaLake are the strongest classes (F1 0.99), with Residential, River, and Highway close behind (F1 0.98) — these have either a very distinctive texture or a clean, separable colour profile in true colour. PermanentCrop is the weakest class by a clear margin (F1 0.94), with Pasture and HerbaceousVegetation next (F1 0.96) — all three are continuous-canopy vegetation classes that look visually similar to each other and to AnnualCrop without the benefit of the extra spectral bands.

Multispectral — Strengths & Weaknesses

Forest, Residential, and SeaLake are classified almost perfectly (F1 0.99–1.00). Pasture and PermanentCrop remain the model's weakest classes (F1 0.97) — notably the same two classes that were weakest for RGB, which is a useful sanity check: it tells us this is a genuine ambiguity in the data (these classes really do look and reflect similarly) rather than a quirk of one particular model.

3.3 Confusion Matrix

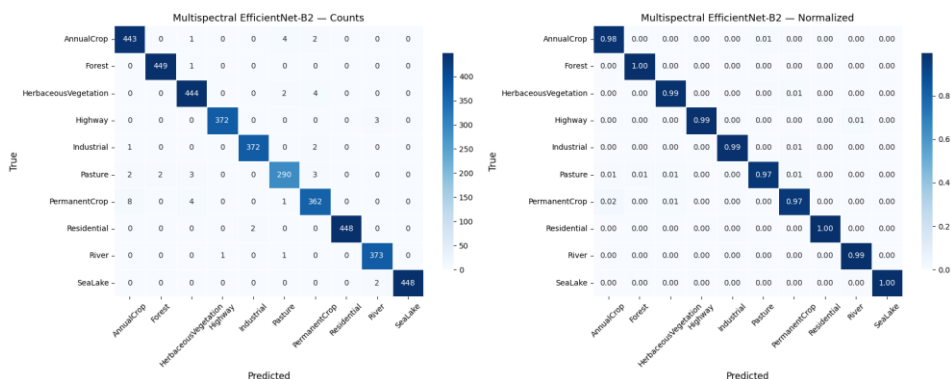


Figure 3.1 — Multispectral model confusion matrix (counts and row-normalized). Almost all the mass sits on the diagonal; the RGB model's matrix follows a similar but visibly noisier pattern, especially around the vegetation/crop classes — available in the notebook, Section 11, for that plot.

3.4 Top Confusion Pairs

Model	True Class	Predicted Class	Count (top-50 errors)
RGB	HerbaceousVegetation	PermanentCrop	11
RGB	AnnualCrop	PermanentCrop	6
RGB	HerbaceousVegetation	Pasture	4
RGB	Highway	AnnualCrop	3
MS	AnnualCrop	Pasture	4
MS	HerbaceousVegetation	PermanentCrop	4
MS	Highway	River	3

Table 3.2 — Most frequent true→predicted error pairs among each model's 50 highest-confidence mistakes.

Why These Classes Are Confused

- HerbaceousVegetation ↔ PermanentCrop / Pasture: all three are continuous green-canopy land covers with similar texture in RGB and overlapping reflectance in the visible bands — the difference is mainly land-management context (wild vs. cultivated vs. grazed) rather than a strong spectral signal, so even the multispectral model retains residual confusion here.
- AnnualCrop → Pasture (multispectral): occurs mainly for early-growth-stage crop patches, where canopy density and NDVI are temporarily close to pasture levels before the crop matures — a known seasonal ambiguity in optical crop monitoring.
- Highway ↔ River / AnnualCrop (RGB): elongated, linear, low-vegetation strips look geometrically similar to roads in true colour; the multispectral model resolves most of this via the strong NIR/SWIR contrast between water/vegetation and bare asphalt, which is why Highway → AnnualCrop confusion disappears for the MS model while a smaller Highway → River confusion remains (both are linear, low-texture, low-vegetation features).

3.5 Confident Misclassifications (Error Deep-Dive)

The most informative errors are confident mistakes — cases the model is sure about but wrong. Grad-CAM overlays on the highest-confidence errors show where the model's attention was focused when it went wrong.

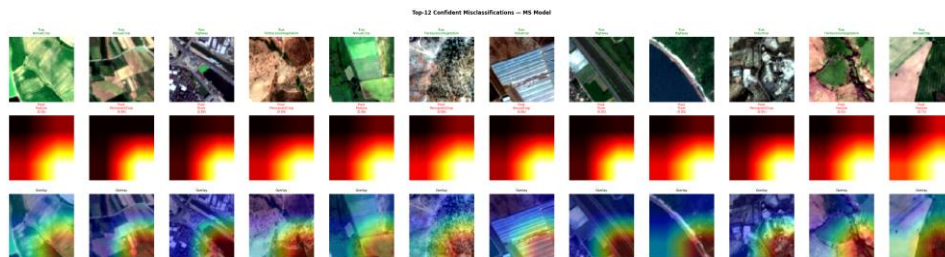


Figure 3.2 — Top confident misclassifications, multispectral model (false-colour NIR-Red-Green composite). Each triplet shows the input, Grad-CAM heatmap, and overlay; the RGB model's equivalent grid is available in the notebook, Section 15 (Misclassification Deep Dive).

3.6 Does Spectral Information Help? — Per-Class Gain

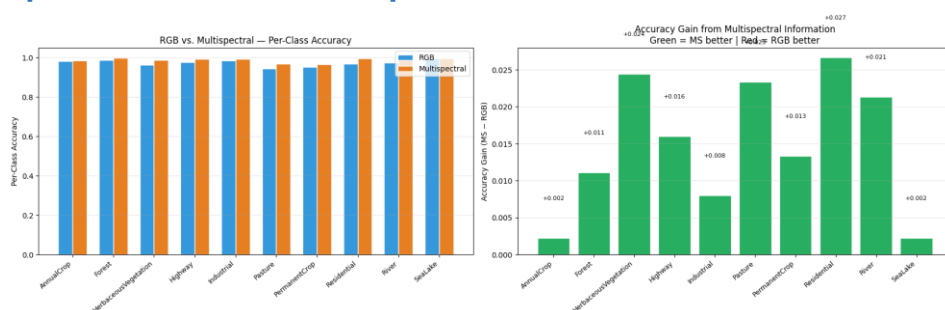


Figure 3.3 — Per-class accuracy, RGB vs. multispectral (left), and the accuracy gain from adding spectral bands (right).

Overall accuracy improves from 97.33% (RGB) to 98.79% (multispectral) — a +1.46 point gain. The gain is not uniform: Residential (+2.67 pts), HerbaceousVegetation (+2.44 pts), Pasture (+2.33 pts), River (+2.13 pts), and Highway (+1.60 pts) see the largest improvements, confirming the hypothesis in the problem statement that spectrally distinct but visually similar classes benefit most from the extra 10 bands.

4. Explainability (Task 2)

4.1 Visual Explainability — Grad-CAM

Grad-CAM is computed on the final convolutional block of EfficientNet-B2: the gradient of the predicted class score with respect to that feature map is global-average-pooled into per-channel weights, which weight the activations to produce a class-discriminative spatial heatmap, upsampled to the input resolution.

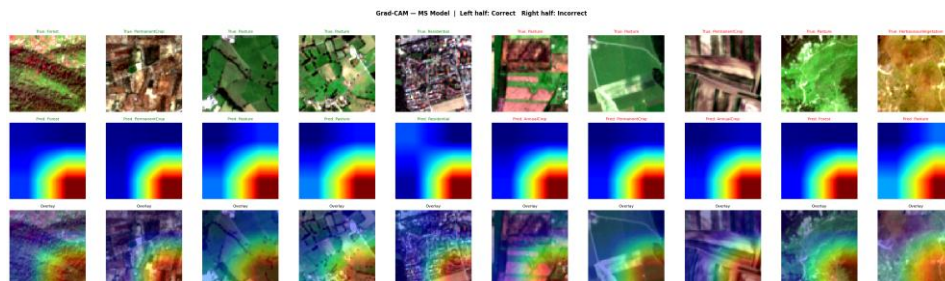


Figure 4.1 — Grad-CAM grid, multispectral model (correct predictions left half, incorrect right half). On correct predictions, attention concentrates on semantically meaningful regions — field texture for crops, footprint edges for Residential/Industrial, shoreline for SeaLake/River — rather than image borders or background, indicating the model has learned genuine land-cover cues rather than spurious shortcuts. The RGB model's grid shows the same pattern and is available in the notebook, Section 12.

4.2 Spectral / Band Importance

Per-band importance is measured by occlusion: each of the 13 bands is zeroed out in turn and the resulting drop in validation accuracy is recorded relative to the unperturbed baseline (98.88% on this 800-image sample).

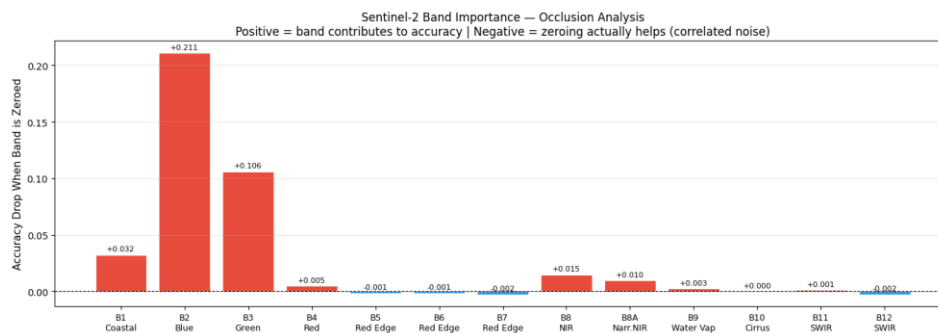


Figure 4.2 — Accuracy drop when each Sentinel-2 band is individually zeroed (occlusion importance).

- Blue (B2, +0.211) and Green (B3, +0.106) cause by far the largest accuracy drops when removed — these are the channels that inherit the strongest, most reliable priors from ImageNet-pretrained RGB weights, so the network leans on them most heavily.
- NIR (B8, +0.015) and Narrow NIR (B8A, +0.010) contribute next most — consistent with NIR's well-established role in vegetation discrimination in remote sensing (high chlorophyll reflectance).
- Red Edge (B5–B7), Cirrus (B10), and several other bands show ~0 or even slightly negative drops — not because they are useless, but because single-band occlusion underestimates the value of correlated bands: when 13 jointly-informative channels are available, the network can partly compensate for the loss of any one of them using its neighbours. This is a known limitation of single-feature occlusion analysis, and is why the model's 98.79% multispectral accuracy still meaningfully exceeds the 97.33% RGB accuracy even though no single extra band shows a large individual drop.

This lines up with the per-class spectral signature plot available in the notebook, Section 14: vegetation classes (Forest, Pasture) show a sharp jump in reflectance from Red into NIR — the classic vegetation "red-edge" — while SeaLake shows the opposite pattern, absorbing rather than reflecting in NIR/SWIR. Both are textbook remote-sensing signatures, which is a good sanity check that the bands the model leans on (Section 4.2 above) are physically meaningful rather than incidental.

4.3 Error & Confusion Summary

Confusion matrices, top error pairs, and confident-misclassification visualizations are presented together in Section 3.3–3.5, alongside the spectral/visual reasoning for each major confusion pair, to keep error analysis and explainability cross-referenced rather than duplicated.

5. Bonus — Environmental Insights (Task 3)

This task asked us to go beyond classification accuracy and pull out real environmental or geospatial insight from the multispectral data. We computed three standard remote-sensing indices directly from raw (un-normalized) Sentinel-2 reflectance values on a 1,000-image validation sample:

Index	Formula	What It Reveals
NDVI	$(\text{NIR} - \text{Red}) / (\text{NIR} + \text{Red})$	Photosynthetically active vegetation density
NDWI	$(\text{Green} - \text{NIR}) / (\text{Green} + \text{NIR})$	Open water bodies and surface moisture
NDBI	$(\text{SWIR1} - \text{NIR}) / (\text{SWIR1} + \text{NIR})$	Built-up / urban surfaces

Table 5.1 — Spectral indices used for environmental insight.

Class	NDVI	NDWI	NDBI
Forest	0.720	-0.578	-0.664
Pasture	0.669	-0.554	-0.573
Highway	0.442	-0.403	-0.347
River	0.430	-0.315	-0.459
HerbaceousVegetation	0.392	-0.381	-0.246

Class	NDVI	NDWI	NDBI
AnnualCrop	0.371	-0.406	-0.296
PermanentCrop	0.361	-0.380	-0.229
Residential	0.347	-0.320	-0.237
Industrial	0.231	-0.218	-0.202
SeaLake	-0.220	0.431	-0.734

Table 5.2 — Mean per-class spectral index values, validation set (n=1,000).

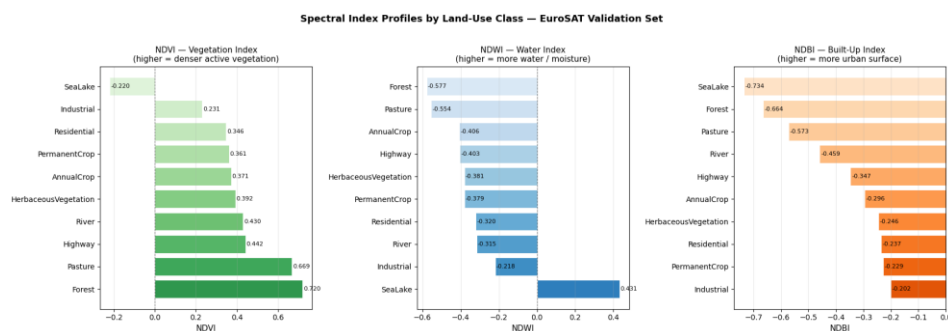


Figure 5.1 — NDVI, NDWI, and NDBI profiles ranked by class. The three indices below are read directly off this chart and Table 5.2. (The notebook, Section 17, also has an NDVI histogram and NDVI-vs-NDWI/NDBI scatter plots for the same data, shown there rather than repeated here.)

5.1 Vegetation Health & Crop Monitoring (NDVI)

Forest (mean NDVI 0.72) and Pasture (0.67) sit clearly in the dense, healthy-vegetation range, while AnnualCrop (0.37) and PermanentCrop (0.36) are about half that and noticeably more variable across images. This makes sense: forests and pastures are continuously green, whereas crop fields pass through bare soil, early growth, and full-canopy stages within a single season, so any one snapshot can land anywhere along that curve. This is exactly the signal real crop-monitoring systems use: a field's NDVI dropping well below its usual seasonal level is a standard early warning for drought stress, disease, or an approaching harvest, often before the change is visible to the human eye.

5.2 Water & Moisture Mapping (NDWI)

NDWI cleanly separates SeaLake (mean +0.43) from every single land class, all of which are negative. That is a sharp, almost binary boundary — it means open water can be picked out directly from the index, without needing a trained classifier at all. The practical use of this is tracking the same water body over time: a shrinking NDWI over a lake or reservoir is a simple, quantitative way to flag drought-driven water loss, which is one of the more visible and policy-relevant signs of regional climate stress.

5.3 Urban / Built-Up Surface Behaviour (NDBI)

Among the built classes, Industrial has the highest NDBI (-0.202, the least negative) and the lowest NDVI — the most "concrete-heavy" signature in the dataset. Residential is close behind on NDBI (-0.237) but noticeably greener on NDVI (0.347 vs. 0.231) — the expected effect of gardens, street trees, and lawns softening an otherwise built environment. Repeatedly mapping NDBI over the same urban fringe over time would show how fast impervious surface is replacing green space, which feeds directly into urban heat-island and city-planning studies.

5.4 Towards Land Suitability & Geo-Planning

Reading NDVI and NDBI together also gives a simple, surprise-us-style use case the problem statement specifically calls out: land suitability. A patch with high NDVI and strongly negative NDBI (Forest, Pasture) is clearly vegetated and ecologically valuable land; a patch with low NDVI and the least-negative NDBI (Industrial) is already built and effectively unavailable; classes in between (AnnualCrop, PermanentCrop, HerbaceousVegetation) sit in a middle zone that is plausible to repurpose for agriculture or controlled urban expansion with the least ecological cost. Running this same two-index check across a region's Sentinel-2 tiles would give planners a first-pass, low-cost shortlist of land to investigate further — instead of manually inspecting every parcel.

5.5 Relevance to Climate Monitoring

All three indices come from freely available Sentinel-2 imagery with a 5-day global revisit, so the same classification-plus-index pipeline used here can, in principle, be run repeatedly over time on any region to build a low-cost, automated monitoring layer:

- NDVI time-series over agricultural land can flag drought stress or yield decline early enough to support intervention, ahead of visible crop damage.
- NDWI time-series over lakes/reservoirs gives an early, quantitative signal of water-resource stress and shrinkage — directly relevant to regions facing climate-driven drought.
- NDBI time-series over urban fringes quantifies the pace of land conversion from vegetated to built-up surface, informing urban-heat-island mitigation and land-use planning.

Combined with the land-use classifier from Task 1, this turns a single satellite pass into both a land-cover label and a quantitative environmental health signal for that pixel — at a scale and frequency that would be impractical with field surveys alone.

6. Bonus — Web Application (Deployment)

Outside the core deliverables, we also wrapped both final models in a small Streamlit web app, so that the classifier is something anyone can actually try, not just something that runs inside a notebook. It loads our saved RGB and multispectral EfficientNet-B2 checkpoints directly and serves live predictions on uploaded imagery.

Live demo: <https://geosnap-pixelpioneers.streamlit.app/>

What It Does

- Automatically detects whether an uploaded file is an RGB image or a 13-band Sentinel-2 file, and routes it to the matching model — no manual mode switch needed.
- Accepts RGB images in JPG / JPEG / PNG / BMP / WEBP, and 13-band Sentinel-2 multispectral images in TIF / TIFF.
- Runs real-time inference with the same trained EfficientNet-B2 checkpoints used throughout this report, and shows the prediction confidence alongside the Top-3 predicted classes.
- Plots a full probability-distribution chart across all 10 classes for the uploaded image.
- For multispectral uploads, computes the environmental indices from Section 5 — NDVI, NDWI, and a Moisture Index — directly from the uploaded scene.
- Lets the user download a prediction report for the uploaded image.

Technology Stack

- Streamlit — web UI and file upload
- PyTorch + EfficientNet-B2 — model loading and inference
- NumPy, Pandas, Matplotlib — index computation and plotting

This was built mainly so we (and anyone reviewing the project) could sanity-check the model on a fresh image in a few seconds, without opening the notebook — and it doubles as a quick demo for the Stage 2 presentation.

7. Conclusion

- Multispectral input improves validation accuracy from 97.33% (RGB) to 98.79% (13-band) — a measurable, reproducible gain from spectral information beyond what a camera can see, concentrated in classes that are visually similar but spectrally distinct (Residential, HerbaceousVegetation, Pasture, River, Highway).
- EfficientNet-B2 with AdamW + cosine warm restarts was selected after a 4-way comparison against ResNet18 baselines and an alternative learning-rate schedule, on the basis of both higher accuracy and lower training cost (Section 2.4).
- Grad-CAM and band-occlusion analysis both indicate the model has learned semantically and spectrally meaningful cues rather than spurious shortcuts, and its remaining confusions (Pasture / PermanentCrop / HerbaceousVegetation) align with genuine, known ambiguity in vegetated land-cover classes.

- NDVI/NDWI/NDBI analysis on the same multispectral data surfaces actionable environmental signals — crop stress, water-body change, urban encroachment, and a simple land-suitability check — directly from the inputs the classifier already uses, at Sentinel-2's global, repeated-revisit scale.
- As a bonus, both final models are deployed in a live Streamlit app (Section 6), so the classifier can be tried on a fresh image in seconds rather than only inside the notebook.

Deliverables

File	Contents
rgb_predictions.csv	Task 1A test-set predictions (img_id, predicted_label)
ms_predictions.csv	Task 1B test-set predictions (img_id, predicted_label)
rgb_predictions_proba.csv / ms_predictions_proba.csv	Per-class softmax probabilities (used for confidence-based error analysis)
efficientnet_b2_rgb_final.pth / efficientnet_b2_ms_final.pth	Self-describing checkpoints: weights + class list + normalization stats
PixelPioneers_GeoSnap.ipynb	Full training, inference, explainability, and bonus-insight pipeline
Technical_Report.docx	This report
Web app (bonus) — URL	Live demo serving both final checkpoints for interactive testing